Experiment No. 09

Aim: - Write a program using Applet:
- to display a message in the Applet.
- for configuring Applets by passing parameters

Software Requirement: - Java Development Kit (JDK), Integrated Development Environment (IDE) (NetBeans), Java Runtime Environment (JRE).

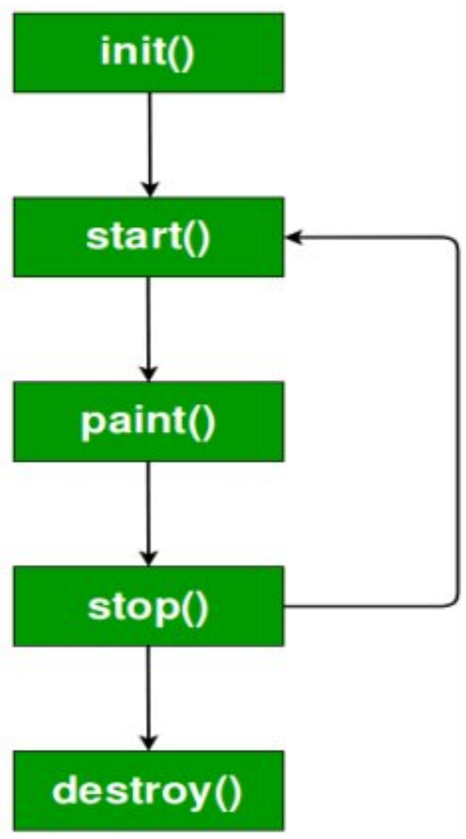
Theory: -

Java Applet

Java Applets was once a very popular feature of web applications. Java Applets were small programs written in Java that ran inside a web browser. A Java Applet is a Java program that runs inside a web browser. An Applet is embedded in an HTML file using <applet> or <objects> tags. Applets are used to make the website more dynamic and entertaining. Applets are executed in a sandbox for security, restricting access to local system resources.

Java Applet Life Cycle

The below diagram demonstrates the life cycle of Java Applet:



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It is important to understand the order in which the various methods shown in the above image are called.

- When an applet begins, the following methods are called, in this sequence:
  - init( )
  - start( )
  - paint( )
- When an applet is terminated, the following sequence of method calls takes place:
  - stop( )
  - destroy( )

Let's look more closely at these methods.

1. init( ): The init( ) method is the first method to be called. This is where you should initialize variables. This method is called only once during the run time of your applet.

2. start( ): The start( ) method is called after init( ). It is also called to restart an applet after it has been stopped.

Note: init( ) is called once i.e. when the first time an applet is loaded whereas start( ) is called each time an applet's HTML document is displayed onscreen. So, if a user leaves a web page and comes back, the applet resumes execution at start( )

3. paint( ): The paint( ) method is called each time an AWT-based applet's output must be redrawn. This situation can occur for several reasons. For example, the window in which the applet is running may be overwritten by another window and then uncovered. Or the applet window may be minimized and then restored.

- paint( ) is also called when the applet begins execution. Whatever the cause, whenever the applet must redraw its output, paint( ) is called.
- The paint( ) method has one parameter of type Graphics. This parameter will contain the graphics context, which describes the graphics environment in which the applet is running. This context is used whenever output to the applet is required.
- paint() is the only method among all the methods mention above (which is parameterized).

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4. stop( ): The stop( ) method is called when a web browser leaves the HTML document containing the applet, when it goes to another page.

For example: When stop( ) is called, the applet is probably running. You should use stop( ) to suspend threads that don't need to run when the applet is not visible. You can restart them when start( ) is called if the user returns to the page.

5. destroy( ): The destroy( ) method is called when the environment determines that your applet needs to be removed completely from memory. At this point, you should free up any resources the applet may be using. The stop () method is always called before destroy ().

Key Packages for Java Applets

- java.applet.Applet: Base class for applets.
- java.awt.Graphics: Used for drawing on the applet screen.
- java.awt: Provides GUI components and event-handling mechanisms.

Advantages of Applet

- It runs inside the browser and works on the Client-side, so it takes less time to respond.
- It is more Secured
- It can be Executed by multi-platforms with any Browsers, i.e., Windows, Mac Os, Linux Os.

Disadvantages of Applet

- A plugin is required at the client browser (User Side) to execute an Applet.

Program and Output:

Save this as MyApplet.java.

```
import java.applet.Applet;
import java.awt.*;

/*
 <applet code="MyApplet" width="300" height="200">
    <param name="message" value="Hello, Welcome to Java Applet!">
```

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```
 </applet>
*/

public class MyApplet extends Applet {
   String message;

   // Initialize the applet
   public void init() {
      message = getParameter("message"); // Retrieve the parameter
      if (message == null) {
         message = "Default Message: Hello, Applet!";
      }
   }

   // Paint method to display message
   public void paint(Graphics g) {
      g.setFont(new Font("Arial", Font.BOLD, 14));
      g.drawString(message, 20, 100);
   }
}
```

Save this as MyApplet.html.

```
<html>
   <body>
      <applet code="MyApplet.class" width="300" height="200">
         <param name="message" value="Hello, Welcome to Java Applet!">
      </applet>
   </body>
</html>
```

Output:

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VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Conclusion

This applet program demonstrates how to display a message and configure an applet dynamically using parameters. By using the <param> tag and getParameter(), applets can be customized for different use cases without changing the Java source code.

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